

PURBANCHAL UNIVERSITY

2025

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 60 /Pass Marks: 24

MCA104/MCA114: Advanced Database Management System (New/Old)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. Compare and contrast lock-based and timestamp-based concurrences control. Check whether the schedule S is conflict serializable or not? 6+6
S: R3(y); R3(z); R1(x); W1(x); W3(y); W3(z); R2(z); R1(y); W1(y); R2(y); W2(y); R2(x); W2(x)
2. How can active databases be used in real-time applications like fraud detection or monitoring systems? What are the challenges and solutions for managing geospatial data in databases? 7+5
- 3(a) What are the different types of join algorithms used in query execution? Explain. 6
- (b) Explain the difference between cost-based and rule-based query optimizer. 6

Group B

Answer SIX questions.

6×6=36

4. Consider the following table:
EMPLOYEES (Employee_ID, First_Name, Last_name, Email, Phone_Number, Start_Date, Salary, Manager_Id, Department)
 - (a) Find the employee with the highest salary.
 - (b) Write a query to group employees by department and find the total no. of employees in each department.
 - (c) Display the details of employees who do not belong to "HR" department.
5. Explain the concept of parallel and distributed databases.
6. What is database security? Explain the challenges of database security.
7. What are the possible reasons of disk failures? How can data be recovered from disk crashes?
8. Explain the tier architecture of DBMS.
9. What is SOAP? Explain X-query and X-path with examples.
10. What is indexing? Briefly explain the types of indexes.
11. What do you mean by undo and redo logging? Define check point in the context of system recovery.

(2)

3. What is log-based recovery technique? Explain immediate update and deferred update techniques with suitable example.

Group B

Answer SIX questions.

6×6=36

4. What is RAID-5 and how does it work? How does RAID improve data redundancy and fault tolerance?
5. Explain architecture of DBMS with a suitable example.
6. What is XQuery, and how is it used to query XML data? What is a schema, and how is it used in semi-structured data?
7. What is the difference between a clustered index and a non-clustered index? How does indexing improve search and retrieval operations in DBMS?
8. What are the emerging database technologies, and how do they differ from traditional database technologies in terms of their features and capabilities?
9. What is rule-based query optimization, and how does it work in database system?
10. Differentiate between authentication and authorization with respect to database security. Explain various levels of database security.

///